Customer Personality Analysis and Segmentation Using Machine Learning in Python

Algorithm

1. Step-by-step Procedure

1. **Data Loading & Cleaning:** Import customer records, handle missing values (e.g., impute missing income with median), and remove extreme outliers in age/income.
2. **Feature Engineering:** Calculate total spending, derive total household children, compute promotional responses, and generate customer tenure/age metrics.
3. **Data Preprocessing:** Encode categorical variables (education, marital status) and scale numerical features using StandardScaler.
4. **Dimensionality Reduction & Clustering:** Apply Principal Component Analysis (PCA) and perform K-Means Clustering using the Elbow Method/Silhouette Score to determine k.
5. **Model Evaluation & Classification:** Analyze segment profiles and train a classifier to predict customer segments and campaign responses for new data.

2. Logic or Method Used

- **Unsupervised Customer Segmentation:** Grouping customers based on historical purchasing behaviors, demographic attributes, and engagement metrics without predefined labels.
- **Feature Aggregation:** Summarizing multi-channel spending and promotional responses into high-level behavioral indicators to simplify cluster identification.

3. Model or Technique Applied

- **Dimensionality Reduction:** Principal Component Analysis (**PCA**) for feature compression and noise reduction.
- **Clustering Model: K-Means Clustering** for segmenting customers into distinct personality profiles.

- **Supervised Classification: Random Forest Classifier** (or **XGBoost**) to classify new/unseen customer records into existing segments.

- ## 4. Process Flow of the System

Plaintext

- `Raw Data Input ➔ Data Preprocessing & Cleaning ➔ Feature Engineering ➔ Standard Scaling`
- `   ➔ PCA Dimensionality Reduction ➔ K-Means Clustering ➔ Profile Labeling & Classification Model`

2. Dataset

1. Source of Data

- **Source:** Kaggle — Customer Personality Analysis Dataset (provided by iFood).

2. Number of Records/Samples

- **Total Records:** 2,240 customer rows.

3. Features or Attributes (Columns)

- **Demographics:** ID, Year_Birth, Education, Marital_Status, Income, Kidhome, Teenhome
- **Product Spending:** MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds
- **Channel Preferences:** NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth
- **Promotion & Engagement:** AcceptedCmp1–AcceptedCmp5, Response, Complain, Dt_Customer
- **Engineered Features:** Age, Total_Spending, Children, Total_Promos

4. Training and Testing Data Split

- **Split Ratio:** 80% Training Data, 20% Testing Data (applied during the supervised classification phase).

5. Data Format or Type

- **File Format:** Comma-Separated Values (.csv).
- **Data Types:** Numerical (int64, float64), Categorical (object), and Datetime (datetime64).